

* Aim:- To determine the flash point of lubricating oil by Pensky Martens apparatus.

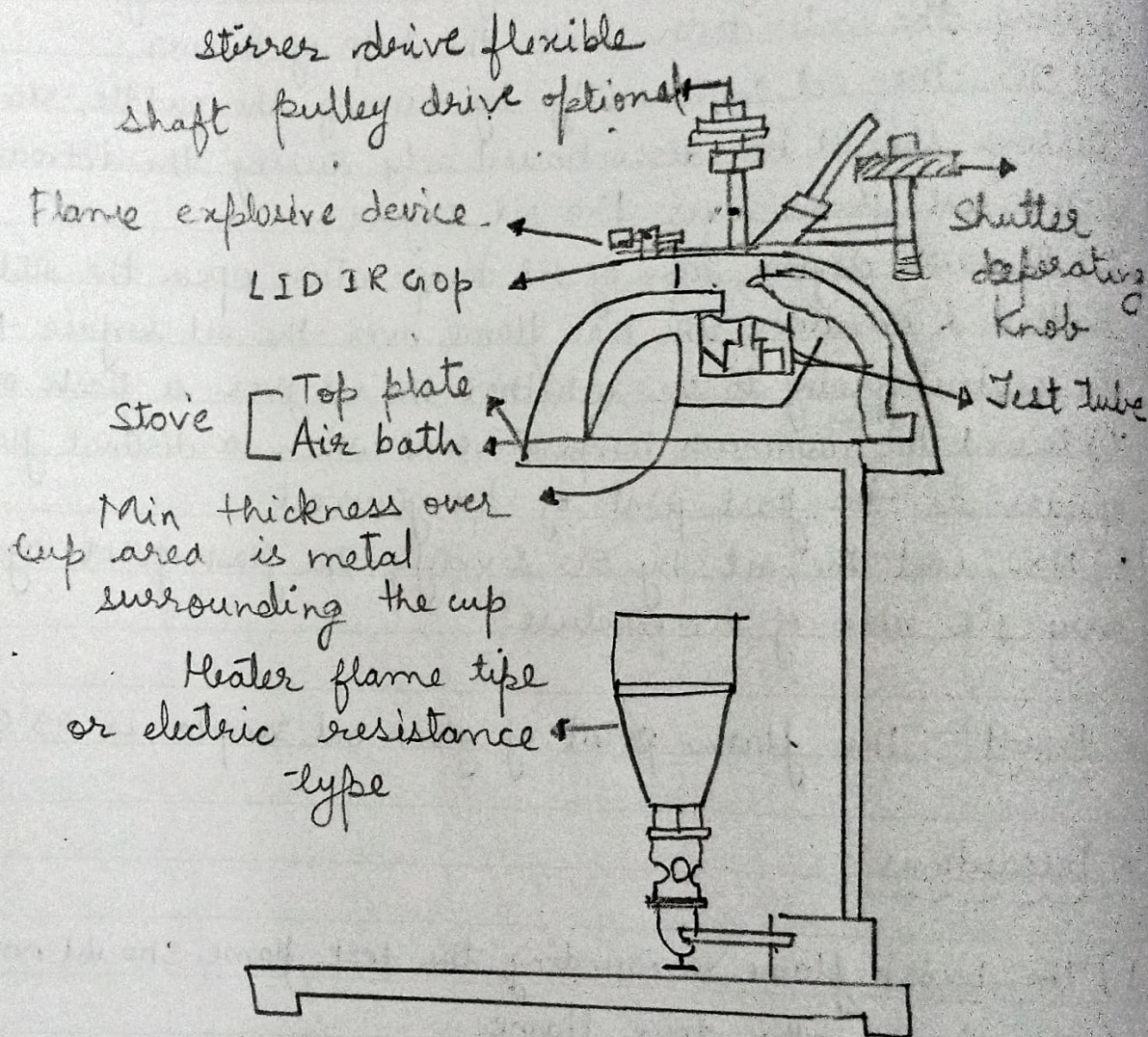
* Articles Required:- Pensky Martens apparatus, thermometers, oil sample & LPG etc.

* Definition:- Flash point of oil is defined as the minimum temperature at which it gives off sufficient vapours, which when mixed with air forms an ignitable mixture & gives a momentary flash of light on the application of a small flame.

* Significance of determination:- The determination of flash point is of great importance in case of lubricating & illuminating oils. If the oils are sufficiently volatile at ordinary temperatures, the issuing vapours will form an explosive mixture with air & cause fire hazards. To ensure safety while using oils, certain minimum temperatures are laid down for illuminating & lubricating purposes, below which it should not give off inflammable vapours.

The flash point of oil shows the extent of more volatile constituents in the oil. The lower the flash point the greater is the amount of more volatile constituents.

The flash point is clearly distinguished from fire point, which is the temperature at which the air space above the oil has sufficient concentration of vapours so that it burns continuously for about five seconds when brought in contact with the flames.



Pensky Martens Apparatus

★ Part of Apparatus:- The Pensky Martens apparatus consists of the following main parts:-

1.) The Oil Cup:- It is a cylindrical vessel, made of brass with a filling mark grooved inside near the top. The cup is covered with a lid.

2.) Lid:- The lid is equipped with following parts:-

(i) Stirrer:- The stirring device is carrying two bladed brass propellers mounted in the centre of the cup.

(ii) Cover Proper:- It is made of brass & is provided with four openings, one for thermometer & the rest for the oxygen entry & exposure of vapours to test flame.

(iii) Shutter:- It is made of brass & is so shaped & mounted on the lid that when in one position the holes are completely closed & when in the other, these openings are completely opened so as to test the material for its flash point. It can be worked by rotating the knob.

(iv) Flame Exposure Device:- The lid is provided with a pilot lamp with such a mechanism that its flame operates simultaneously with the shutter. When the shutter is in the 'open' position the tip of the flame is lowered down.

(v) Heating Device:- The oil cup is placed in another cup of slightly bigger dimensions. Air is present between the two cups thus providing an air bath which is heated electrically.

* Procedure:-

- 1.) The oil cup and its accessories are cleaned & dried before the test is performed.
- 2.) Fill the oil cup with already dehydrated oil sample to be tested up to the filling mark. (A groove cut inside the oil cup). Then place the cup in the heating bath.
- 3.) Replace the cover of the oil cup.
- 4.) Insert the thermometer into the thermometer socket, provided on the cover proper.
- 5.) Heat the oil cup from below with burner with constant slow stirring.
- 6.) After every 10°C rise of temperature upto 100°C & thereafter every 5°C rise of temperature, open the shutter by operating knob & introduce test flame over the oil surface through the central opening to see whether the oil gives flash or not.
- 7.) When the test flame causes a distinct flash in the interior of the cup, note temperature which represents the flash point.
- 8.) Place the oil cup in the tray & allow it to cool down by 10 to 15°C start heating the sample again & test for the flash point after every 1°C rise in temperature till flash observed.
- 9.) Again cooldown the oil cup by 2 to 5°C then check flash point by every 1°C rise of temperature till flash observed.
- 10.) Repeat the last step until two consecutive concurrent temperature.

Observation table :-

S.NO	temperature	flash point
1.	32°C	No flash
2.	42°C	No flash
3.	52°C	No flash
4.	62°C	No flash
5.	72°C	No flash
6.	82°C	No flash
7.	92°C	No flash
8.	102°C	No flash
9.	112°C	No flash
10.	122°C	flash observed

observation table :-

S.NO	temperature	flash point
1.	112°	No flash
2.	113°	No flash
3.	114°	No flash
4.	115°	No flash
5.	116°	No flash
6.	117°	No flash
7.	118°	No flash
8.	119°	No flash
9.	120°	No flash
10.	121°	flash observed

* Result:- The flash point of given oil sample is 121°C

* Precautions:-

- 1.) The bluish flame surrounding the test flame should not be confused with true flame.
- 2.) The traces of solvent used for cleaning the apparatus should be removed completely.

* Aim:- To determine Steam Emulsification Numbers of an oil sample.

* Articles Required:- Thermometers, oil sample, SEN apparatus & LPG

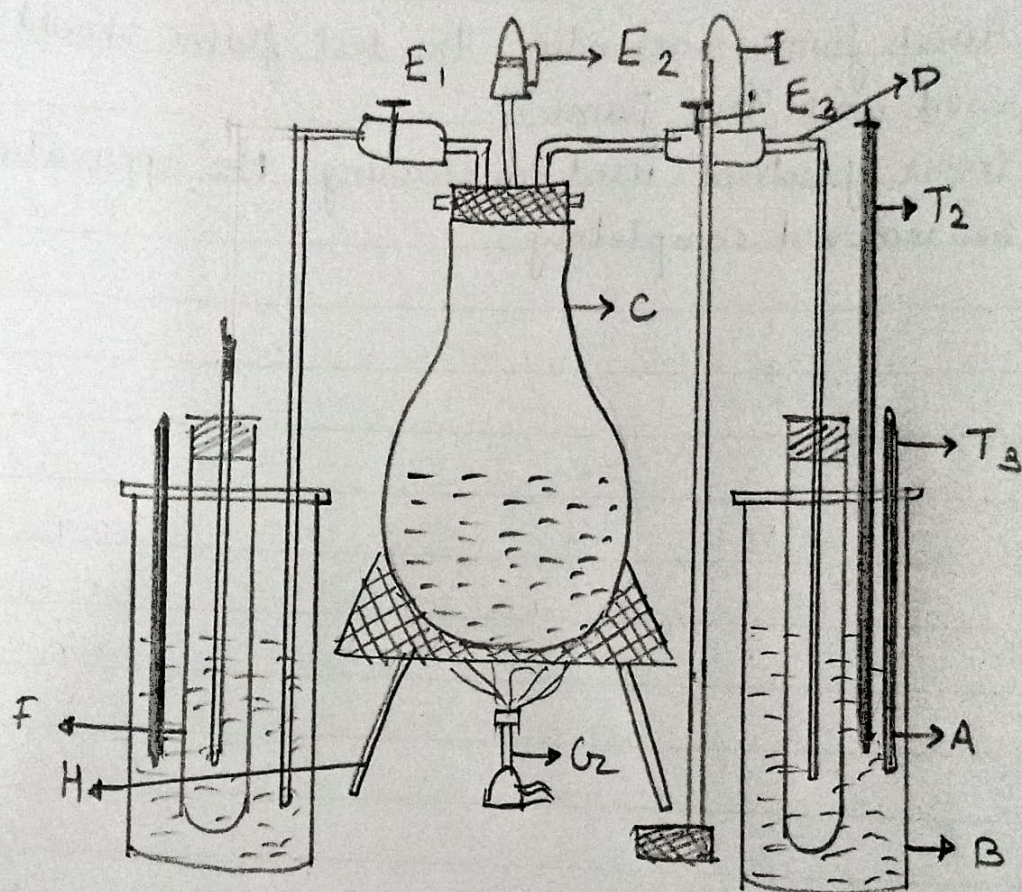
* Definition:- The time in seconds elapsed during the separation of 20 ml of oil from approximately double of its emulsion with water is reported as steam emulsification number.

* Significance of Determination:- Certain lubricating oils mix intimately with water and form emulsion, which then has a tendency to collect dirt, grit particles & other foreign matters. Thus, solid particles are introduced into the lubricated parts & cause abrasion & wearing out of the machinery.

The tendency of oil to emulsify increases with the time, it is used as lubricant. Good lubricating oil is one which does not form emulsion under lubricating conditions & even if it forms, it should break off very quickly.

The tendency of emulsion is determined by A.S.T.M. emulsion test. An emulsion of oil & water is made under standard conditions & then allowed to break. The time in seconds required for the complete separation of water from emulsion is reported as SEN.

In circulating oiling systems, the high SEN is particularly undesirable because sticky emulsions & sludge interfere with the circulation of oil. But in certain steam cylinder oils, the formation of stable emulsion is desirable. In steam turbines,



- A - Graduated Test Tube
- B - Emulsifying Bath
- Gr - Burner
- T_1, T_2, T_3 - Thermometer
- H - Tripod stand
- I - Clamp stand
- C - Steam generator
- E_1, E_2, E_3 - Pinch cocks
- F - separating Bath
- D - Delivery tube

the oils should separate quickly so that it can be re-circulated for use.

SEN or RE (Rate of Emulsion separation) is directly proportional to the emulsification of tested oil.

To establish a basis for comparison, a definite value 100 is assigned to an oil which under condition of test separates completely from emulsion in one minute. The rate is 33 cc per second or 20 cc per minute. It was found after experimentation that the results were pertinent & relative through a wide range of tests. The value assigned is denoted RE or resistance to emulsion.

The RE of any oil may be found by substituting in following formula:-

$$RE = \frac{\text{CC of separated oil} \times 5}{\text{Number of minutes}}$$

* Parts of apparatus:-

- (i) Steam generator 'C':- It is a round bottomed flask heated from below with the burner.
- (ii) Test Tube:- A 2.5 by 20 cm test tube 'D' used as a sample tube graduated (in some) from zero to 50 ml.
- (iii) An emulsifying bath 'B':- It is a glass battery jar of 3 litre capacity.
- (iv) Separating bath 'F':- It is a glass battery jar of 3 litre capacity.

(vi) Delivery Tube 'D':- Steam from 'C' is led into the sample by means of a delivery tube, the other end of which is cut at an angle of 30°C .

* Procedure:-

- 1) Take approximately 20 ml of the oil sample in a test tube. Adjust the thread mark on the test tube up to 40 ml.
- 2) Place the test tube into the beaker kept in tray (half filled with water).
- 3) Now, check that steam generator is filled with water (nearly half of its capacity).
- 4) Fill the other beaker $\frac{3}{4}^{\text{th}}$ with water.
- 5) Now, heat this beaker as well as steam generator from below with the help of burners separately.
- 6) The temperature of water in beaker should be less than 100°C (as soon as some vapours form on the water, stop heating).
- 7) When a continuous steam is coming out from the delivery tube, introduce it to the test tube containing oil sample. (nearly up to the bottom).
- 8) Continue passing steam until the content of the test tube becomes double (reaches up to the thread mark).
- 9) Immediately remove the steam delivery tube & transfer the test tube into the beaker (containing hot water) along with the cover.
- 10) Note down the temperature in seconds required for complete separation of water & oil as SEN.

Observation Table:-

S-NO .	SEN
1 .	49
2 .	50
3 .	50

11.) Now, transfer the contents of the test tube into the emulsion bottle kept in the tray.

12.) Repeat the exercise thrice.

* Result:- Steam Emulsification No. of the given oil is 50.

* Precautions:-

1.) Steaming rate should be controlled as described.

2.) After emulsification, the test tube should be transferred rapidly to the separating bath.

* Aim :- To describe the Aniline point of the given oil sample

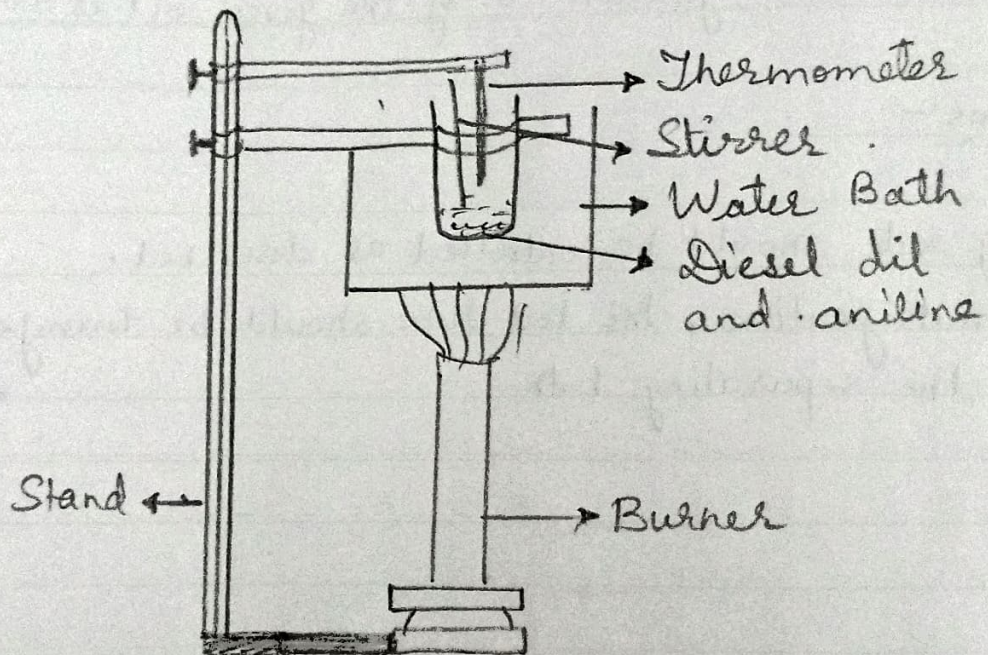
* Articles Required :- Graduated test tube, Aniline point apparatus, Thermometer, water bath, & stirrer etc.

* Definition :- Aniline point may be defined as the lowest temperature at which the oil is completely miscible with an equal volume of aniline.

* Significance of Determination :- Determination of aniline point of fuel oils is very informative.

If petroleum or mineral fuel oils contain a higher percentage of straight chain hydrocarbons, it produces a rattling noise in the internal combustion engine called knocking. This knocking is minimum, if the percentage of branched chain hydrocarbons & or aromatic hydrocarbons is higher. Determination of aniline point gives information about the type of hydrocarbon content of the fuel oil.

If, the aniline point is lower, the proportion of aromatic hydrocarbons is higher & the oil will have less or no tendency of knocking in the engine. Aromatic hydrocarbons present in lubricants have the tendency to dissolve rubber or synthetic rubber packings, sealings etc., therefore, aniline point of oil gives indication regarding the deterioration, dissolution of rubber sealing, packing etc. in contact with lubricating oil. For good lubricating oil, the aromatic



Laboratory Aniline point apparatus

hydrocarbon content should be low. The higher the aniline point, the lower is the percentage of aromatic hydrocarbons in the fuel or lubricating oil & is safer to use in contact with rubber parts. Aniline point gives valuable information about the presence of type hydrocarbon in the oil.

* Parts of apparatus:- The aniline point apparatus consists of:-

(i) Test Tube:- It is to hold the sample. It is made of heat resistant glass & is approximately 25 mm in diameter & 150 mm in length. It is fitted with a cork bored to help the thermometer centrally & provided with another hole through which stirrer can operate freely. This tube is suspended in a jacket, made of similar glass material. The test tube along with jacket is kept immersed in a heating or cooling bath.

(ii) The stirrer:- It is made of metal or glass rod bent at the bottom into a ring about 15 mm in diameter at right angle to the main axis.

(iii) The thermometer:- is suspended in the tube through a central hole in the cork.

* Procedure:-

1) Equal volume of pure & anhydrous aniline & oil are taken in the test tube. (two layers are observed).

Observation :-

Volume of oil taken = V ml .

Volume of aniline taken = V ml .

Observation table :-

S. NO	Temperature at which cloudiness appears in homogeneous solution ($^{\circ}\text{C}$)
1 .	65°C
2 .	68°C
3 .	67°C .

lowest value - 65°C .

- 2.) Water bath is heated & test tube containing oil & aniline is placed in it.
- 3.) The contents of test tube are stirred (by up & down motion of stirrer) till a homogeneous solution is obtained.
- 4.) Thermometer is placed in the test tube & temperature is recorded.
- 5.) The test tube is then placed in the cooling jacket of the aniline point apparatus.
- 6.) Initially, cloudiness appears in the solution & then two layers are obtained.
- 7.) The temperature at which cloudiness appears is noted.
- 8.) The lowest temperature at which cloudiness appears is reported as aniline point.
- 9.) Procedure is repeated thrice, till two concordant results are obtained.

* Result & conclusion :- The aniline point of the given lubricating oil is 65°C .

- 1.) If the aniline point is low, the percentage of aromatic hydrocarbons is higher & the oil should not be used in contact with rubber.
- 2.) If the aniline point is higher, the percentage of aromatic hydrocarbons is lower & oil is suitable for use in contact with rubber.

* Precautions :-

- 1.) Pure aniline should be used.

Teacher's Signature _____

- 2) Stirring should be done by up & down motion of stirrer with care. No part of the mixture should be allowed to splash out.
- 3) Aniline point should be recorded with care & experiment should be repeated at least two times for consistency.

* Aim:- Determination of drop point of a grease sample.

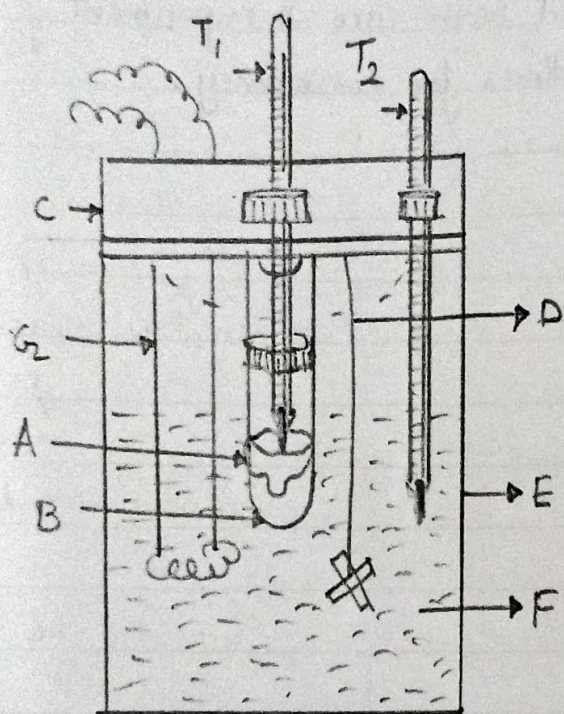
* Articles Required:- Drop point apparatus, water bath, semi-solid lubricant (grease), and thermometer, etc.

* Significance of Determination:- Drop point is the temperature at which the grease passes from semi-solid state to liquid state. It determines the upper limit of temperature up to which grease can function as a satisfactory lubricant.

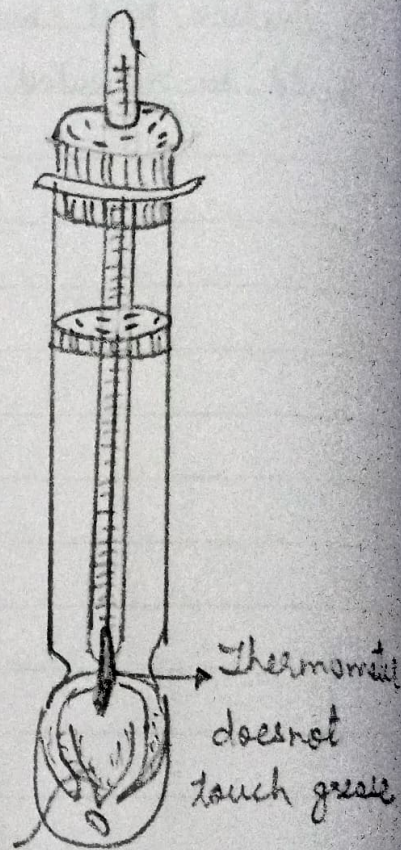
* Parts of apparatus:- It consists of a metal cup having an opening of a standard dimension at the bottom. The cup is fitted in a glass jar (boiling tube) with a tightly fitting lid. A thermometer is fixed in the cup such that the bulb of the thermometer is just above the surface of the grease sample under test. The entire assembly is held in position in a glass beaker containing water which is also provided with a thermometer & a stirrer.

* Procedure:-

- 1) Place the grease sample in the cup & insert the thermometer in place.
- 2) Fix the cup along with thermometer into a boiling tube containing cotton. Now, place the boiling tube into a beaker containing water.



Dropping point apparatus



- A - Chromium plated brass cup
- C - cork
- E - glass tube
- G - Heating coil
- T₂ - Bath thermometer
- B - Pyrex glass test tube
- D - stirrer
- F - Transparent oil
- T₁ - Test thermometer

Observation table:-

S. NO	Drop point ($^{\circ}\text{C}$)
1.	77°C
2.	78°C

3) Heat the beaker from the below with burner slowly. As the temperature increases the grease sample melts & the first drop emerges out of the opening at the bottom of the metal cup.

4) Record the temperature at which the first drop falls down. This temperature is noted as drop point.

5) Now, remove the burner.

6) Take out small amount of hot water (20-25 ml) & mix cold water.

7) The temperature of water should fall down by 10°C .

8) Replace the test tube into the beaker & repeat the experiment 2-3 times, till two readings are similar.

* Result: - The drop point of given grease sample is 77°C .

* Precautions: -

1) Pure aniline should be used.

2) Drop point should be recorded with care & experiment repeated at least two times for consistency.